Editorial

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Climate research in the Global South

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The consequences of weather and climate extremes are felt disproportionately by some of the most disadvantaged populations of the world. Gaps in data and research perpetuate these deeply rooted inequalities and must be overcome.



(https://pixabay.com/photos/biking-street-flood-city-bike-5680458/).

Variable weather, powerful natural disasters, and increased climate extremes associated with anthropogenic climate change are a global phenomenon, but the impacts and implications are felt most severely by disadvantaged communities. A contrast can be seen between the regions that have most benefited from fossil fuel exploitation—the Global North, responsible for some 92% of excess global carbon emissions[1]—and those that are most affected by the consequences of those emissions—the Global South[2,3]. And while the Global South is disproportionately impacted by climate-related natural disasters, it is largely underrepresented in climate research and published literature[4,5]. The latest IPCC report[6] does show some positive change, with the largest representation of authors from the Global South to date, but the majority of research (62% of IPCC authors) still comes from the Global North. With the goal to support robust climate research from the Global South, as an equal and essential counterpart to research from the Global North, *Nature Communications* and *Communications Earth & Environment*, launched a collection on "Weather and Climate Extremes", with a focus on climate research from, and about, the Global South.

To stress the importance of the Global South in climate research, and in recognition of the geographical disparities in the drivers and consequences of climate change, the World Climate Research Program held its once-in-a-decade "Open Science Conference" in October 2023 in Kigali, Rwanda. The conference brought together almost 1500 climate and environment experts from all over the world, with 32% of the participants coming from Africa and 26% from other regions of the Global South. A recurrent discussion topic during the conference was the growing collaboration between research communities from the Global North and South and how, though important, it is not sufficient. Beyond collaboration, there is a need for the global community to give space for autonomy in climate research and not to tokenize communities in the Global South. Those most impacted by climate change must be able to define and prioritize their own research agenda.

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In 2023 catastrophic floods hit East Africa following an unprecedented sequence of five consecutive seasons of drought[7] and in 2024 deadly floods occurred in West Africa. These floods resulted in numerous casualties and damage and are an ongoing threat to food security in the region. One of the main obstacles hindering effective early warning for weather-related natural disasters in Africa is the scarcity of weather radars covering the African continent, along with their operational problems, such as the low accuracy of the delivered data[8,9]. 2024 also brought devastating floods in eastern Spain, in southern Brazil/northern Uruguay, and in Southeast and South Asia. Climate research in the Global South in particular is further hindered by a lack of long, reliable observational records[10]. Given the large internal variability that can mask the anthropogenic climate change signal, long observational records are necessary to identify and understand climate extremes. At least 30 years of continuous observations are required to define a climatic period and allow for robust conclusions, but unfortunately this is yet not realistic in many regions of the Global South, including Latin America and the Caribbean region[11]. Recent developments in state-of-the-art climate models, advanced statistical methods, and machine learning techniques[12,13] have started to fill this observational gap[14] through production of physically plausible storylines[15] or thousands of years of surrogate weather conditions[16]. But climate model development remains slow in several regions of the Global South owing to limited understanding of certain physical processes because of lack of sufficient dedicated studies and good quality datasets, consequently leading to poor model performance[9]. Climate researchers in the Global South further often struggle with limited access to data and computing

power[9], a lack of sufficient funding for research, and language barriers as many local and regional studies are written in languages other than English and are thus less likely to be included in the IPCC assessments[11].

To support climate research in the Global South, the World Climate Research Program has launched an academy to provide a knowledge hub dedicated to the promotion and advancement of lifelong learning opportunities and global equity in climate science training. It is hoped that initiatives like this will facilitate global networking and equity amongst climate scientists, and ultimately connect them so that they can collaborate on research projects. Moving forward, the international climate community should be encouraged to convene regularly in regions of the Global South that have traditionally been overlooked, and where colonialism and its manifestation in helicopter research (the practice of scientists from wealthy nations visiting lower-income countries, doing research and publishing results without the involvement of local researchers)[17,18] have shaped the research landscape for centuries. More meetings in these regions will help give a voice to a currently underrepresented, but crucial, pool of researchers. It is now well accepted that for people and communities to adapt to climate change, local and indigenous knowledge must be heard and taken into account in research design and policy making, without marginalization[19,20].

As a journal, we have a responsibility to recognize and overcome biases in the editorial and peer review process[21] and to actively work on better representing the global research community and their priorities within our author, reviewer, and editor pools. In the "Weather and Climate Extremes" collection we sought out research from around the world, with a special focus on the Global South. The result is a collection that encompasses the unprecedented modern anthropogenic drought in Central Brazil[22]; identification of areas most at risk of increased drought like South and Central America, Australia, the Mediterranean, and southern Africa[23]; inequalities in exposure to compound drought-heatwaves under climate change, showing that low-income regions experienced a twice as fast increase in the frequency of such compound events compared to high-income regions[24]; the increased risk of heat-related mortality in Europe, South-East Asia and the Latin American Pacific coast over the past 20 years and its continued projected increase under higher levels of global warming[25]; windows of opportunity for the prediction of seasonal climate extremes, such as the 2022 floods in Pakistan[26]; the anthropogenic and natural drivers of intensified flash droughts in South Asia[27]; and other works on the drivers and impacts of extreme weather and climate.

An editorial or a collection will certainly not solve the problem of the deeply rooted inequities between the Global South and Global North, but we aspire to signal our willingness to all of those interested in helping us take some small, but persistent, steps towards a more equitable climate research landscape.

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